Provability as failed refutation: A is provable from Gamma iff Gamma plus not-A is inconsistent

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the source immediately follows with a prob asking for the sibling equivalence (Gamma |- not A iff Gamma + {A} inconsistent) - not part of this card; all axiom citations carry their part tags [prp]/[ded]/[ptn] verbatim.

Editorial note. The label prop:prov-incons names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Monotonicity of derivability. Same part, \ollabel{prop:monotonicity} (source line 845). Line 967: Gamma |- A lifts to Gamma + {not A} |- A.
- Reflexivity of derivability. Same part, \ollabel{prop:reflexivity} (source line 836). Line 968: the added hypothesis not A is derivable from the extended set.
- Propositional axiom not-A -> (A -> falsum). Propositional-logic part, axiom ax:lnot2. Line 969, and again implicitly in the converse at line 975.
- Modus ponens admissibility for derivability. This part, \ollabel{prop:mp} (source line 638). Lines 970 and 977-979: three applications closing both directions.
- Axiom falsum-elimination (not-A -> falsum) -> not-not-A. Propositional-logic part, axiom ax:lfalse2. Line 976 in the converse direction.
- Double negation elimination as an axiom. Propositional-logic part, axiom ax:dne. Line 978: from not-not A to A.

- Deduction theorem. Same chapter, the deduction theorem (source line 656). Line 974: inconsistency of Gamma + {not A} becomes Gamma |- not A -> falsum.

Source reference. Original proposition prop-mp: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 639–639. Prerequisite pointer invoked at proof line(s) 971; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ax-dne: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 507–507. Prerequisite pointer invoked at proof line(s) 979; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ax-lfalse2: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 506–506. Prerequisite pointer invoked at proof line(s) 977; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ax-lnot2: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 503–503. Prerequisite pointer invoked at proof line(s) 970; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-monotonicity: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 846–846. Prerequisite pointer invoked at proof line(s) 968; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-reflexivity: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 837–837. Prerequisite pointer invoked at proof line(s) 969; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. $\Gamma \vdash !A$ *iff* $\Gamma \cup \{\neg !A\}$ *is inconsistent.*

Proof. First suppose $\Gamma \vdash !A$. Then $\Gamma \cup \{\neg !A\} \vdash !A$ by prop-monotonicity. $\Gamma \cup \{\neg !A\} \vdash \neg !A$ by prop-reflexivity. We also have $\vdash \neg !A \to (!A \to \bot)$ by ax-lnot2. So by two applications of prop-mp, we have $\Gamma \cup \{\neg !A\} \vdash \bot$.

Now assume $\Gamma \cup \{\neg !A\}$ is inconsistent, i.e., $\Gamma \cup \{\neg !A\} \vdash \bot$. By the deduction theorem, $\Gamma \vdash \neg !A \to \bot$. $\Gamma \vdash (\neg !A \to \bot) \to \neg\neg !A$ by ax-lfalse2, so $\Gamma \vdash \neg\neg !A$ by prop-mp. Since $\Gamma \vdash \neg\neg !A \to !A$ (ax-dne), we have $\Gamma \vdash !A$ by prop-mp again. □